

Unimodular Distribution Certificates and Exact Reflection Torsion for Cyclotomic Polylogarithms

Integral normal forms, arbitrary characteristic, and finite-order jets

Research report prepared for Vladimir Reshetnikov

A continuation of the ProveIt Polylogarithms manuscript

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Abstract

The characteristic-zero distribution normal form in the ProveIt manuscript leaves integral lattices, modular specialization, and reflection torsion as separate questions. We give a fixed basis consisting of original grid symbols and a unit-determinant minor made from unscaled prime distribution rows. Consequently the complete weighted relation matrix is unimodularly equivalent to an identity block and zeros over $\mathbb{Z}[A_p : p \mid q]$, and remains so after arbitrary commutative-ring base change. A division-free recurrence supplies exact polynomial certificates and explicit coefficient bounds.

Reflection behaves differently. A weighted Anderson resolution reduces its Tate cohomology to a finite Koszul complex over the coefficient ring modulo two. This gives the complete Smith torsion for all integer prime weights. For weights in $\mathbb{Z}[u]/(u^L)$, the answer is controlled by one minimum binary valuation: the torsion multiplicity is $v 2^{r_*-1}$, where r_* counts the odd prime factors and also counts 2 precisely when $4 \mid q$. We also exhibit a reflected jet module whose torsion-free part is not free over the jet ring.

Specializing the normal forms to $A_p = p^{1-s}$ produces meromorphic polylogarithm identities for every complex order, including explicit level-12 and level-30 formulas and pole-cancelled identities for all order derivatives at $s = 1$. The package includes independent exact certificate replay, direct Smith computations, and separately labelled numerical diagnostics. The ordinary-distribution and resolution antecedents of Kubert, Anderson, and Ouyang are retained; no worldwide priority or numerical period-independence claim is made.

Scope. All theorem and proposition statements below have ordinary mathematical proofs. The executable checks test finite instances and implementations; they are not proof-assistant formalizations. The remaining mixed-depth S_6 identity is not proved in this report.

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1 Research target and relation to the existing manuscript

1.1 What is being completed

The input manuscript *Polylogarithms and their Arithmetic Bridges* and its distribution-jets report are read at the repository snapshot

28357e8ca63dd78327db91d9be239d75e4462879.

The report’s universal conductor normal form is stated over a characteristic-zero field containing character values. Its polynomial weights may be specialized in any algebra over that field. The manuscript explicitly leaves integral Smith forms, torsion, and integral certificate arithmetic for further work [1, 2].

The distinction is substantive. A Fourier basis change divides by group orders. Thus a characteristic-zero proof of rank does not, by itself, prove saturation over $\mathbb{Z}$, preserve a basis in characteristic two, or identify the torsion introduced by the equations $e_{-x} = \pm e_x$. Conversely, the vanishing of a determinant attached to a particular primitive-grid coordinate map need not be a failure of the complete presentation. This report treats the complete prime-row presentation, not an unspecified numerical lattice of periods.

The principal results are the original-symbol unit-minor theorem (Theorem 2.1), the reflection Koszul reduction (Theorem 6.2), and the exact one-variable jet formula (Theorem 7.1). Together they settle these integral and modular questions for the stated module, including its reflected scalar and one-variable-jet specializations. The polylogarithm consequences are proved in Section 8.

1.2 Classical antecedents and the novelty boundary

Integral freeness of the ordinary universal distribution is classical [3]. Ouyang develops a broader universal norm distribution, proves freeness, and gives an Anderson-type resolution [4, Propositions 4.1–4.2 and Theorem 5.1]. The ordinary sign-cohomology calculation is also classical; Ouyang’s spectral-sequence treatment records the result and its antecedents [5, Theorem 7.1].

Accordingly, neither ordinary-distribution freeness nor the general idea of an Anderson resolution is presented as a new discovery. The contribution here is an explicit, self-contained treatment of the manuscript’s exact weighted presentation: a raw-row unit minor, an original-grid-symbol compiler, the binary Koszul data for reflection, and a closed minimum-valuation formula for arbitrary integral finite jets. These statements are proved rather than inferred from an analogy with the classical theory. Their precise extent of prior appearance in the broader norm-distribution literature is not settled by the literature check made for this report.

1.3 Three different assertions

A basis of a formal distribution quotient means independence modulo the *declared distribution rows*. It does not mean that the evaluated complex numbers are linearly independent. An exact integer certificate proves a relation from those rows; a small floating-point residual is only a diagnostic. Finally, a torsion class in a reflected formal quotient is not torsion of a complex period: evaluation in a characteristic-zero vector space annihilates such a class.

These distinctions will be used throughout. In particular, none of the rank or Smith calculations below proves irrationality of a Clausen value, independence of Dirichlet L -values, or a minimal multiple-polylogarithm depth.

2 The weighted module and a fixed original-symbol basis

Let $q \geq 2$ and write

$$q = \prod_{p \in P} p^{e_p}, \quad X_q = (q^{-1}\mathbb{Z})/\mathbb{Z}, \quad R = \mathbb{Z}[A_p : p \in P].$$

All rings are commutative with identity. Put

$$M_q = \bigoplus_{x \in X_q} R e_x.$$

For $p \in P$ and $x \in X_{q/p}$ define

$$r_{p,x} = \sum_{\substack{y \in X_q \\ py=x}} e_y - A_p e_x. \quad (2.1)$$

Let $\mathcal{I}_q$ be the R -span of every such row, and set $\mathcal{U}_q = M_q/\mathcal{I}_q$. There are q columns and $m_q = \sum_{p|q} q/p$ row rows. The symbol e_0 is retained. For polylogarithmic evaluation it means $\text{Li}_s(1) = \zeta(s)$; for Hurwitz-zeta evaluation it means $\zeta(s, 1)$, not $\zeta(s, 0)$.

If $d \mid q$, put $A_d = \prod_p A_p^{v_p(d)}$. The divisor row is defined in the same way, with $dy = x$ and weight A_d . The identity

$$r_{ab,x} = \sum_{ay=x} r_{b,y} + A_b r_{a,x} \quad (ab \mid q) \quad (2.2)$$

proves by induction that prime rows generate all divisor rows. The multiplicativity of composite weights is essential here.

2.1 Canonical local selectors

Write each point uniquely in its primary decomposition

$$x = \sum_{p \in P} \frac{a_p(x)}{p^{e_p}}, \quad 0 \leq a_p(x) < p^{e_p}. \quad (2.3)$$

If $x = a/q$, the local numerator is

$$a_p(x) \equiv a (q/p^{e_p})^{-1} \pmod{p^{e_p}}.$$

Define the local forbidden and retained sets

$$T_p = \{1, \dots, p^{e_p-1}\}, \quad B_p = \{0\} \cup \{p^{e_p-1} + 1, \dots, p^{e_p} - 1\}. \quad (2.4)$$

The proposed basis set is

$$\mathcal{B}_q = \{x \in X_q : a_p(x) \in B_p \text{ for every } p \mid q\}. \quad (2.5)$$

It has $\varphi(q)$ elements and contains zero. The set T_p selects one *nonzero* lift from each multiplication-by- p fiber on the local grid. For a nonzero parent numerator $b < p^{e_p-1}$ it selects b ; for parent zero it selects p^{e_p-1} .

Let $h_p(x)$ be the exponent of the exact order of the p -component of x , with $h_p(x) = 0$ for a zero component. Write

$$H(x) = \sum_p h_p(x), \quad b(x) = \#\{p : a_p(x) \in T_p\}.$$

For $x \notin \mathcal{B}_q$, let $p(x)$ be its least forbidden prime and choose one original row

$$g_x = r_{p(x), p(x)x}. \quad (2.6)$$

Order all grid points lexicographically by $(H(x), b(x), a)$, where $x = a/q$ and $0 \leq a < q$.

Theorem 2.1 (Original-symbol unimodularity). *The classes $[e_b]$, $b \in \mathcal{B}_q$, form an R -basis of $\mathcal{U}_q$. The selected raw rows g_x , $x \notin \mathcal{B}_q$, form an R -basis of $\mathcal{I}_q$. In the indicated order, their submatrix on the nonbasis columns is unit lower triangular. In particular, its determinant is identically one.*

The complete $m_q \times q$ raw relation matrix is equivalent, by invertible row and column operations over R , to an identity block of size $q - \varphi(q)$ and zeros. All these statements survive arbitrary base change $R \rightarrow T$. In particular,

$$\mathcal{U}_q \otimes_R T \simeq T^{\varphi(q)}, \quad (\mathcal{U}_q / Re_0) \otimes_R T \simeq T^{\varphi(q)-1}.$$

Example 2.2 (Small bases). We identify a/q with its numerator in the following display:

q	$q\mathcal{B}_q \pmod{q}$
3	0, 2
4	0, 3
6	0, 4
12	0, 5, 8, 9
15	0, 1, 4, 6, 7, 9, 10, 12
30	0, 2, 8, 12, 14, 18, 20, 24.

These are not primitive-root grids. Including lower-conductor symbols is exactly what permits one basis to work for every weight without exceptional denominators.

3 Proof of the unit-minor theorem

3.1 Triangularity without division

Fix a nonbasis point x , and put $p = p(x)$. In the row g_x , the point x has coefficient one. Its p -component is nonzero, and multiplication by p lowers its exact local order. Thus $px \neq x$ and the weight term cannot change that pivot coefficient.

Every other point in g_x precedes x . Indeed, the other solutions of $py = px$ have the same components away from p . Among those of the same p -order as x , the unique forbidden lift was x itself; all the other lifts are retained at p . Their height is unchanged and their forbidden count decreases by one. A possible zero local lift has lower height. Finally px has height $H(x) - 1$: multiplication by p preserves the orders at every other prime, even though it may change their forbidden status. Height takes priority over forbidden count in the ordering.

Two different selected points cannot select the same row. For a fixed prime and parent there is exactly one forbidden lift in its local fiber; the other local components are uniquely fixed by the parent. Therefore there are exactly $q - \varphi(q)$ distinct selected rows, and their nonbasis submatrix is unit lower triangular.

Let K_q be their span. Successive unit-pivot elimination gives a direct sum

$$M_q = K_q \oplus \bigoplus_{b \in \mathcal{B}_q} Re_b. \quad (3.1)$$

What remains is to prove that the unselected raw rows belong to K_q . This step is necessary: a unit minor alone gives a lower bound on the rank of the relation module, not its full description.

3.2 The complete generic rank

We give the characteristic-zero argument in full so that the integral descent does not rely on unseen relation matrices. Extend scalars to a field F containing the fraction field of R and all character values of $(\mathbb{Z}/q\mathbb{Z})^\times$. Split the grid according to exact denominator:

$$F[X_q] = \bigoplus_{m|q} F[(\mathbb{Z}/m\mathbb{Z})^\times].$$

Use $(\mathbb{Z}/1\mathbb{Z})^\times$ as a one-element group. For a primitive character χ of conductor $f \mid q$ and every $f \mid m \mid q$, put

$$v_{m,\chi} = \sum_{a \in (\mathbb{Z}/m\mathbb{Z})^\times} \chi(a) e_{a/m}.$$

The finite character transforms are invertible over F .

Fourier-transform the *sources* of all p -rows, separately on their exact-denominator strata. For $f \mid m$ and $pm \mid q$, the resulting row is

$$v_{pm,\chi} - \lambda_{p,m,\chi} v_{m,\chi}, \quad \lambda_{p,m,\chi} = \begin{cases} A_p, & p \mid m, \\ A_p - \chi(p), & p \nmid m. \end{cases} \quad (3.2)$$

When $p \mid m$, every preimage has denominator pm . When $p \nmid m$, one preimage has denominator m ; its character coefficient is $\chi(p)\chi(b)$, since its numerator satisfies $pb \equiv a \pmod{m}$. All remaining preimages have denominator pm . This proves (3.2). Because both source and target transforms are invertible, these are all transformed rows, not a selected collection of identities.

Define

$$E_{m,\chi} = \prod_{p|f} A_p^{v_p(m) - v_p(f)} \prod_{\substack{p|m \\ p \nmid f}} A_p^{v_p(m) - 1} (A_p - \chi(p)). \quad (3.3)$$

The products of edge factors are independent of the order in which primes are adjoined. Every $v_{m,\chi}$ reduces to $E_{m,\chi} v_{f,\chi}$. Conversely, the map $v_{m,\chi} \mapsto E_{m,\chi}$ annihilates every edge and sends $v_{f,\chi}$ to one. The character block therefore has dimension exactly one. There is one such block for each character modulo q , giving

$$\dim_F(F[X_q]/F\mathcal{I}_q) = \varphi(q). \quad (3.4)$$

This is the conductor argument of the earlier report, now used only where the Fourier transform is legitimate [2].

3.3 Integral descent and all-ring specialization

The selected rows are independent and have the full generic rank $q - \varphi(q)$. Hence $FK_q = F\mathcal{I}_q$. For any row r , its class in M_q/K_q becomes zero over F . But (3.1) identifies this quotient with a free module over the integral domain R . It injects into its scalar extension to F . Thus the class of r was already zero: $\mathcal{I}_q = K_q$.

This proves the basis statements. To reduce the entire matrix, put the selected rows first, subtract their polynomial combinations from every remaining row, and use the invertible triangular pivot block to clear the nonbasis columns. A final column operation clears the retained columns of the selected rows. Every operation is invertible over R ; no integer, weight, or character value is divided out. The result is an identity block and zeros.

An identity between these polynomial matrices remains an identity under any homomorphism $R \rightarrow T$. The decomposition, normal forms, and unit minor consequently survive arbitrary specialization, including rings with zero divisors and fields of characteristic dividing q or $\varphi(q)$. Since $0 \in \mathcal{B}_q$, the anchored statement follows by deleting one basis vector. This completes the proof of Theorem 2.1.

4 Division-free certificates, size bounds, and unreflected jets

Write

$$[e_x] = \sum_{b \in \mathcal{B}_q} P_{x,b}(\mathbf{A})[e_b], \quad P_{x,b} \in \mathbb{Z}[\mathbf{A}].$$

The proof gives the explicit recurrence

$$\text{NF}(x) = \begin{cases} e_x, & x \in \mathcal{B}_q, \\ A_p \text{NF}(px) - \sum_{j=1}^{p-1} \text{NF}(x + j/p), & x \notin \mathcal{B}_q, p = p(x). \end{cases} \quad (4.1)$$

Every argument on the second line precedes x . Thus this is a terminating compiler over the polynomial ring itself, not a specialization-dependent Gaussian elimination.

Proposition 4.1 (Degree and coefficient bounds). *For every x, b and $p \mid q$,*

$$\deg_{A_p} P_{x,b} \leq h_p(x).$$

Let $r = |P|$, let $P_{\max}$ be the largest prime divisor of q , and put $W(x) = (r+1)H(x) + b(x)$. If the coefficient norm sums absolute values over all basis coordinates and all monomials, then

$$\|\text{NF}(x)\|_1 \leq P_{\max}^{W(x)}. \quad (4.2)$$

In particular each coordinate has at most $\prod_p (h_p(x) + 1) \leq \prod_p (e_p + 1)$ possible monomials.

Proof. A factor A_p is introduced only on the branch $x \mapsto px$, which lowers h_p by one. All other local order exponents are unchanged. Sibling lifts do not increase any local order. Induction in the well-founded point order proves the multidegree bound.

Every recursion edge lowers W by at least one. At equal height, the forbidden count drops; when height drops, a possible increase of at most r in the forbidden count is outweighed by the factor $r+1$. There are at most $p \leq P_{\max}$ summands in a recursion step, including the weighted parent. Multiplication by a variable does not increase the coefficient norm. Induction proves (4.2). $\square$

These estimates are deliberately uniform rather than sharp. Given the factorization of q , straightforward sparse arithmetic uses at most $O(qP_{\max}\varphi(q)\tau(q)r)$ integer coefficient operations under this monomial bound, where $\tau(q) = \prod_p (e_p + 1)$. This is a polynomial-in-grid-size bound, not an assertion of optimal complexity or of efficient factorization of a binary-encoded input integer.

4.1 Independent replay

A certificate records the basis, the selected original rows, and every polynomial $P_{x,b}$. An independent verifier can check it without re-running the discovery recurrence:

1. Reconstruct each raw row directly from its prime and parent.

2. Verify that the selected nonbasis block is unit lower triangular.
3. Back-substitute $e_x - \sum_b P_{x,b} e_b$ through that block and require an exactly zero polynomial remainder, for every x .
4. Independently require every raw row to vanish under the delivered normal-form map.

The delivered verifier follows this route and does not import the normal-form generator. Altering a coefficient, the advertised basis, or a selected parent is rejected by the corruption controls.

Corollary 4.2 (Unreflected Smith factors and jets). *For every integer specialization of the prime weights, all nonzero Smith factors of the complete raw matrix are one. Its rank is $q - \varphi(q)$ over every field after the corresponding specialization.*

For $B_L = \mathbb{Z}[u]/(u^L)$, $L \geq 1$, and arbitrary weights $A_p(u) \in B_L$, the quotient is free of rank $\varphi(q)$ over B_L . Expanded in the coefficient basis $1, u, \dots, u^{L-1}$, the raw integer matrix has $L(q - \varphi(q))$ nonzero Smith factors, all equal to one.

Proof. Specialize the polynomial matrix equivalence from Theorem 2.1. For jets, multiplication by each entry is represented in the coefficient basis. An invertible matrix over B_L induces an invertible integer matrix on the underlying free abelian group. Expanding the identity block gives the stated result. $\square$

The same argument works for any finite free coefficient algebra, including multivariate truncated polynomial algebras. It concerns ordinary coefficients of formal jets; dividing derivatives by factorials is unnecessary and may be illegitimate in positive characteristic.

5 A weighted Anderson resolution with an elementary filtration proof

The reflection calculation requires relations among the raw relations, not just their rank. We provide an explicit resolution. This is a weighted version of the classical Anderson construction [4]; the exactness proof below is given for the present point-grid convention.

For $S \subseteq P$, put $d_S = \prod_{p \in S} p$, and set

$$\mathcal{L}^{-k} = \bigoplus_{\substack{S \subseteq P \\ |S|=k}} R[X_{q/d_S}]. \quad (5.1)$$

Denote a generator by $[x, S]$. With the primes ordered increasingly, define

$$d[x, S] = \sum_{p \in S} (-1)^{\#\{\ell \in S: \ell < p\}} \left(\sum_{py=x} [y, S \setminus \{p\}] - A_p[x, S \setminus \{p\}] \right). \quad (5.2)$$

Here the preimages lie in $X_{q/d_{S \setminus \{p\}}}$. The augmentation $\mathcal{L}^0 \rightarrow \mathcal{U}_q$ is the defining quotient map.

Theorem 5.1 (Exact weighted resolution). *The augmented complex*

$$0 \longrightarrow \mathcal{L}^{-|P|} \longrightarrow \dots \longrightarrow \mathcal{L}^{-1} \longrightarrow \mathcal{L}^0 \longrightarrow \mathcal{U}_q \longrightarrow 0$$

is exact and remains exact after arbitrary base change of R . It is equivariant under $J[x, S] = [-x, S]$.

Proof. The unsigned operators in distinct prime directions commute: the two iterated preimage sums give the same solutions of $p\ell y = x$, and inclusion and the scalar weight terms commute with them. The alternating signs in (5.2) therefore give $d^2 = 0$. The augmentation has cokernel zero and kernel $\mathcal{I}_q$ by definition. Equivariance is immediate.

Filter the complex by

$$F(x, S) = \sum_p h_p(x) + |S|.$$

Every differential term has filtration at most that of its source. For a prime $p \in S$, the preimages of highest local order have p -height $h_p(x) + 1$. They preserve the filtration. A possible lower local lift, and the scalar term $-A_p[x, S \setminus \{p\}]$, lower it. In the associated graded complex, record

$$\alpha_p = h_p(x) + \mathbf{1}_{p \in S}, \quad 0 \leq \alpha_p \leq e_p.$$

The leading differential preserves the whole tuple α .

There is a harmless prime-to- p twist on the other primary components of a preimage. Remove it in the summand indexed by S by replacing x_p with

$$\left(\prod_{\ell \in S \setminus \{p\}} \ell \right)^{-1} x_p. \quad (5.3)$$

The multiplier is a unit on the p -primary group. If the differential removes $\ell \neq p$, its preimage operation multiplies x_p by ℓ^{-1} , while the coordinate change in (5.3) gains the factor ℓ . These cancel. In the p -direction the multiplier is unchanged. Thus the graded complex for a fixed α is a tensor product of independent local complexes.

For $\alpha_p = 0$, the local complex is one copy of R in degree zero. For $\alpha_p \geq 1$, it is a two-term complex from the free module on points of exact order $p^{\alpha_p - 1}$ to that on points of exact order p^{α_p} , sending a point to its highest-order preimage sum. At $\alpha_p = 1$, the source is the zero point and its image is the sum of the $p - 1$ nonzero p -torsion points. At larger α_p , each fiber has p points. In either case the fibers are disjoint and nonempty, so choosing one point per fiber gives a unit-pivot splitting. The map is split injective with free cokernel.

Each local complex is consequently a direct sum of a contractible identity-map complex and a free module in degree zero. Their tensor product has no negative-degree homology. The filtration is finite, so successively lifting a cycle's highest filtered part proves that the original complex has no negative-degree homology either. This argument uses only unit-pivot splittings and therefore works over every coefficient ring. Its degree-zero homology is precisely $\mathcal{U}_q$. $\square$

Lemma 5.2 (Nonfixed support is preserved). *In (5.2), a generator with $2x \neq 0$ has no nonzero coefficient on a generator $[y, T]$ with $2y = 0$.*

Proof. The weight term has the same point x . A fixed preimage y would give $2x = p(2y) = 0$, a contradiction. Thus neither type of term can land on a fixed point. $\square$

This elementary support fact is the reason the reflection calculation below collapses exactly; it cannot be replaced by an unsupported assertion that a spectral sequence has no higher differentials.

6 Reflection torsion as a binary Koszul calculation

Let B be a commutative ring whose underlying abelian group is free, let $a_p \in B$ be the specialized weights, and write $U_B = \mathcal{U}_q \otimes_R B$. We allow B to have infinite $\mathbb{Z}$ -rank for now. Reflection is the

involution $Je_x = e_{-x}$. Define the two *coinvariant* quotients

$$C_+(U_B) = U_B/(J-1)U_B, \quad C_-(U_B) = U_B/(J+1)U_B. \quad (6.1)$$

They are not the eigensublattices $\ker(J \mp 1)$. Over a ring in which two is invertible this distinction disappears after applying the usual projectors; over $\mathbb{Z}$ it is essential.

6.1 The two-periodic complex

For a free abelian group U with involution J , use the complex whose differential from even degree is $J-1$ and from odd degree is $J+1$. Its cohomology is denoted $\hat{H}^n(J, U)$; in particular

$$\hat{H}^0(J, U) = \frac{\ker(J-1)}{(J+1)U}, \quad \hat{H}^1(J, U) = \frac{\ker(J+1)}{(J-1)U}.$$

This is Tate cohomology for the two-element group, but these explicit formulas suffice for its use here.

Lemma 6.1 (Only elementary two-torsion). *For a free abelian U ,*

$$\mathrm{Tor}_{\mathbb{Z}} C_+(U) \simeq \hat{H}^1(J, U), \quad \mathrm{Tor}_{\mathbb{Z}} C_-(U) \simeq \hat{H}^0(J, U).$$

Both torsion groups are killed by two.

Proof. If $n[v] = 0$ in $U/(J-1)U$, apply $J+1$ to a representative equation. Since U has no integer torsion, $(J+1)v = 0$. Then $(J-1)v = -2v$, so $2[v] = 0$. Conversely, every element of $\ker(J+1)$ gives such a two-torsion class. This identifies the torsion with $\hat{H}^1$. Replacing J by $-J$ gives the other assertion. $\square$

Put $\bar{B} = B/2B$ and define

$$r_* = \#\{p \mid q : p \text{ odd}\} + \mathbf{1}_{4|q}. \quad (6.2)$$

The list of binary parameters is

$$\mathbf{f} = (1 - \bar{a}_p : p \mid q, p \text{ odd}), \quad \text{with } \bar{a}_2(1 - \bar{a}_2) \text{ appended if } 4 \mid q. \quad (6.3)$$

Let $K_{\bar{B}}(\mathbf{f})$ be the homological Koszul complex: $K_k = \bigwedge^k \bar{B}^{r_*}$ and the differential contracts a basis vector in direction i by multiplication by f_i .

Theorem 6.2 (Fixed-point Koszul reduction). *For every integer n , there is an isomorphism of $\bar{B}$ -modules*

$$\hat{H}^n(J, U_B) \simeq \bigoplus_{\substack{0 \leq k \leq r_* \\ k \equiv n \pmod{2}}} H_k(K_{\bar{B}}(\mathbf{f})). \quad (6.4)$$

The formula includes $q = 2$, when the parameter list is empty.

Proof. Apply the two-periodic complex to each term of the exact resolution in Theorem 5.1, after base change to B . Use the usual sign on one differential so that the total differential squares to zero. The horizontal range is finite. Computing horizontal homology first therefore identifies the total cohomology with $\hat{H}^*(J, U_B)$; this is also obtained by a finite cycle-lifting argument from the augmentation. No unbounded-filtration convergence assertion is needed.

In each vertical column, separate the point basis into fixed points and two-element orbits. Construct a subcomplex D as follows. In even vertical degree include all nonfixed point coordinates

and twice each fixed coordinate. In odd vertical degree include the entire column. A nonfixed orbit has an acyclic two-periodic complex. A fixed coordinate in D has alternating modules $2B, B$ and differential $2 : B \rightarrow 2B$, which is an isomorphism onto its target. Thus every vertical column of D is acyclic. The subcomplex is horizontally stable: by Lemma 5.2, nonfixed source coordinates never produce fixed coordinates, and twice a fixed coordinate still has even fixed coefficients after taking the differential. Since the horizontal range is finite, the total complex of D is acyclic.

The quotient double complex is now particularly simple. Odd vertical degrees are zero; even vertical degrees are the free $\overline{B}$ -module on the fixed points of the corresponding grid. All vertical differentials vanish. Its total cohomology is the direct sum, with the appropriate degree parities, of the homology of the horizontal fixed-point complex.

A grid X_m has just the fixed point 0 if m is odd, and has fixed points $0, 1/2$ if m is even. In an odd-prime direction, each fixed point has exactly one fixed preimage, so the induced map is multiplication by $1 - \overline{a}_p$. These give the odd-prime Koszul factors.

It remains to examine the direction 2 when q is even. If $v_2(q) = 1$, the fixed-point map is

$$\overline{B} \longrightarrow \overline{B}^2, \quad 1 \longmapsto \begin{pmatrix} 1 - \overline{a}_2 & \\ & 1 \end{pmatrix}.$$

It is split injective with one free cokernel coordinate; its contractible summand may be removed. Thus it contributes no Koszul parameter. If $v_2(q) \geq 2$, the map on fixed points is

$$\begin{pmatrix} 1 - \overline{a}_2 & 0 \\ 1 & \overline{a}_2 \end{pmatrix}. \quad (6.5)$$

The first column comes from the source zero, whose two fixed preimages are 0 and $1/2$. The source $1/2$ has only nonfixed preimages, so its fixed contribution is the weight term. Swapping rows and using the unit entry 1 transforms (6.5) into $\text{diag}(1, \overline{a}_2(1 - \overline{a}_2))$ over $\overline{B}$. Removing the identity-map summand leaves exactly the additional Koszul parameter in (6.3).

The other directions commute with these operations. The fixed-point complex is therefore homotopy equivalent to $K_{\overline{B}}(\mathbf{f})$. In total degree n , a homological degree k term contributes precisely when $n + k$ is even. This proves (6.4). $\square$

6.2 Complete scalar Smith torsion

Theorem 6.3 (All integer prime weights). *Let $q > 2$ and specialize each A_p to an arbitrary integer a_p . Then, for either sign,*

$$C_{\pm}(U_{\mathbb{Z}}) \simeq \mathbb{Z}^{\varphi(q)/2} \oplus (\mathbb{Z}/2\mathbb{Z})^{h_q(\mathbf{a})}, \quad (6.6)$$

where

$$h_q(\mathbf{a}) = \begin{cases} 0, & \text{some odd } p \mid q \text{ has } a_p \text{ even,} \\ 2^{r_*-1}, & \text{every odd-prime weight is odd.} \end{cases} \quad (6.7)$$

For $q = 2$, the two quotients are $C_+ = \mathbb{Z}$ and $C_- = \mathbb{Z}/2\mathbb{Z}$.

Proof. Over $\mathbb{F}_2$, the parameter $\overline{a}_2(1 - \overline{a}_2)$ is always zero. If an odd-prime parameter is one, the Koszul complex has a contractible factor and all its homology vanishes. Otherwise every parameter is zero, so $H_k = \mathbb{F}_2^{\binom{r_*}{k}}$. For $q > 2$ we have $r_* \geq 1$, and the sum of binomial coefficients of either parity is 2^{r_*-1} . Theorem 6.2 and Lemma 6.1 prove the torsion formula.

To compute the free ranks without dividing an integral lattice, tensor with $\mathbb{Q}$. The trace of J on the quotient equals its alternating trace on the finite resolution. A point-permutation module has

trace equal to its number of fixed points. For odd q this alternating sum is $(1 - 1)^{|P|}$. If $q = 2m$ with m odd, it is $(2 - 1)(1 - 1)^{\omega(m)}$. If $4 \mid q$, it has the factor $2 - 2$. For every $q > 2$ the trace is therefore zero. Since the total dimension is $\varphi(q)$ and an involution is diagonalizable over $\mathbb{Q}$, its two rational eigenspaces each have dimension $\varphi(q)/2$. This is also the free rank of either coinvariant quotient. The classification of finitely generated abelian groups proves (6.6). At $q = 2$, the unreflected quotient is generated by e_0 and reflection is the identity. $\square$

This result determines the complete nonzero Smith factors of the *stacked* distribution-and-reflection matrix: there are h_q factors equal to two and $q - \varphi(q)/2 - h_q$ factors equal to one. Its remaining $\varphi(q)/2$ column directions are free. In characteristic two the reflected quotient has dimension $\varphi(q)/2 + h_q$; in every odd characteristic it has dimension $\varphi(q)/2$. The unreflected rank never changes.

Example 6.4 (A torsion jump with constant unreflected rank). For $q = 3$, the single raw relation is

$$e_{1/3} + e_{2/3} = (a_3 - 1)e_0.$$

At $a_3 = 1$, reflection is diagonal with signs $+1, -1$ in the basis $e_0, e_{2/3}$, and both coinvariant quotients are $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. At $a_3 = 0$, both coinvariant quotients are $\mathbb{Z}$. The unreflected lattice is $\mathbb{Z}^2$ in both cases. For $q = 60$, $r_* = 3$; the torsion multiplicity is four when both a_3 and a_5 are odd, and zero otherwise, regardless of a_2 .

The effective prime count is not optional. When $q = 2m$ with m odd, the canonical bases for m and $2m$ are the same rational points. The natural inclusion identifies the two modules after adjoining A_2 , and respects reflection: all odd-prime fibers of a point in X_m remain inside X_m . Thus a lone factor of two must not double the ordinary sign-cohomology multiplicity. For ordinary weights the theorem recovers the classical result with this conductor convention [5].

6.3 A universal cohomology module

Corollary 6.5 (Before weight specialization). *Over $R = \mathbb{Z}[A_p : p \mid q]$, Tate cohomology in odd degree vanishes, while in even degree it is*

$$R/(2, 1 - A_p \text{ (} p \text{ odd)}, A_2(1 - A_2) \text{ if } 4 \mid q).$$

A prime variable A_2 with $v_2(q) = 1$ is not constrained in this quotient.

Proof. The binary parameters form a regular sequence in the polynomial ring $R/2R$: the odd-prime linear factors eliminate distinct variables, and, when present, $A_2(1 - A_2)$ is a nonzero monic polynomial in the remaining variable. Its Koszul homology is concentrated in degree zero and equals the quotient by the parameter ideal. Theorem 6.2 applies because R is free as an abelian group. $\square$

The extra cohomology after scalar specialization is thus a real base-change phenomenon. It cannot be read off by simply specializing the already-computed universal cohomology module.

7 An exact Smith formula for arbitrary finite integral jets

Let

$$B_L = \mathbb{Z}[u]/(u^L), \quad L \geq 1, \quad a_p(u) \in B_L,$$

and reduce coefficients modulo two. In $\overline{B}_L = \mathbb{F}_2[u]/(u^L)$, form the parameters (6.3). For an element f of this ring, define

$$\nu(f) = \min\{j : [u^j]f \neq 0\}, \quad \nu(0) = L, \quad v = \min_{f \in \mathbf{f}} \nu(f).$$

For $q > 2$, the list is nonempty.

Theorem 7.1 (Minimum-valuation jet formula). *For $q > 2$, the two reflected jet quotients have the following Smith decomposition as abelian groups:*

$$C_{\pm}(U_{B_L}) \simeq \mathbb{Z}^{L\varphi(q)/2} \oplus (\mathbb{Z}/2\mathbb{Z})^{v2^{r_*-1}}. \quad (7.1)$$

More precisely, their integer-torsion submodules are each isomorphic, as B_L -modules, to

$$(B_L/(2, u^v))^{2^{r_*-1}}. \quad (7.2)$$

The direct-sum decomposition in (7.1) is a statement over $\mathbb{Z}$, not a claim that the torsion-free part is free over B_L .

Proof. In the principal ideal ring $\overline{B}_L$, choose a parameter of least valuation. It is a unit times u^v unless every parameter is zero, which is the same argument with $v = L$. A change of basis in the free module defining the Koszul complex transforms the parameter list into $(u^v, 0, \dots, 0)$: first divide by its unit factor, then subtract its multiples from the other entries. All these are invertible operations over $\overline{B}_L$.

The complex is now the tensor product of the two-term complex given by multiplication by u^v and $r_* - 1$ zero-differential exterior factors. Both the kernel and the cokernel of multiplication by u^v are isomorphic to $\overline{B}_L/(u^v)$; the kernel is the ideal $u^{L-v}\overline{B}_L$. Consequently

$$H_k(K_{\overline{B}_L}(\mathbf{f})) \simeq (\overline{B}_L/(u^v))^{\binom{r_*}{k}}.$$

Summing either degree parity in Theorem 6.2 proves (7.2). Each copy has $\mathbb{F}_2$ -dimension v , so Lemma 6.1 gives the torsion multiplicity in (7.1).

The unreflected underlying lattice has rank $L\varphi(q)$. The alternating fixed-point trace used in Theorem 6.3 is multiplied by L , and still vanishes. Thus each rational eigenspace has dimension $L\varphi(q)/2$. This proves the free rank and completes the Smith calculation. $\square$

Example 7.2 (Five jet coefficients at level fifteen). Take $L = 5$, $a_3(u) = 1 + u^2$, and $a_5(u) = 1 + 2u + u^3$. The binary parameters are u^2, u^3 , so $v = 2$ and $r_* = 2$. Therefore

$$C_{\pm}(U_{B_5}) \simeq \mathbb{Z}^{20} \oplus (\mathbb{Z}/2\mathbb{Z})^4.$$

The constant jets $a_3 = a_5 = 1$ would instead give ten two-torsion factors. The constant terms of the weights agree; their higher binary coefficients change the integral reflection structure.

Example 7.3 (The two-adic jet parameter matters). At $q = 12$, $L = 4$, let $a_2(u) = 1 + u$ and $a_3(u) = 1 + u^3$. The two parameters are u^3 and $u + u^2$, not simply $1 - a_3$ and $1 - a_2$. Thus $v = 1$, $r_* = 2$, and

$$C_{\pm}(U_{B_4}) \simeq \mathbb{Z}^8 \oplus (\mathbb{Z}/2\mathbb{Z})^2.$$

Although the parity of a *scalar* two-prime weight does not affect Theorem 6.3, a nonconstant two-prime jet can affect Theorem 7.1.

7.1 Why the free abelian part need not be jet-free

Take $q = 3$, $L = 2$, and $a_3 = 1 + u$. In the unreflected basis $e = e_0$, $f = e_{2/3}$, reflection satisfies

$$Je = e, \quad Jf = ue - f.$$

Hence

$$C_+(U_{B_2}) = B_2e \oplus B_2f / \langle ue - 2f \rangle. \quad (7.3)$$

As an abelian group this has free generators e, f and a torsion generator uf of order two. In its torsion-free quotient, multiplication by u sends e to $2f$ and f to zero. This is not a free rank-one B_2 -module: the image of u in a free rank-one module is a primitive rank-one sublattice, whereas here the image is $2\mathbb{Z}f$. Thus a claim of B_L -freeness for the reflected torsion-free part would be false. By contrast, $C_- = B_2f \oplus (B_2/(2, u))e$ in this example. Both signs have the same abelian Smith factors, but different module extensions over the jet ring.

7.2 Multivariate jets and arithmetic weights

For a finite free $\mathbb{Z}$ -algebra B of rank L , the Koszul theorem remains an exact answer. If $q > 2$ and $h_k = \dim_{\mathbb{F}_2} H_k(K_{B/2B}(\mathbf{f}))$, then

$$\mathrm{Tor}_{\mathbb{Z}} C_+ \simeq (\mathbb{Z}/2\mathbb{Z})^{\sum_{k \text{ odd}} h_k}, \quad \mathrm{Tor}_{\mathbb{Z}} C_- \simeq (\mathbb{Z}/2\mathbb{Z})^{\sum_{k \text{ even}} h_k}.$$

The two sums are equal: the Euler characteristic of the Koszul complex is $L(1-1)^{r*} = 0$. A closed minimum-valuation description is special to the one-variable principal ideal ring; it must not be transferred unchanged to a multivariate Artinian ring.

For actual positive integral polylogarithm orders, $A_p = p^{1-n}$ usually belongs to $\mathbb{Z}[1/q]$, rather than $\mathbb{Z}$. The corresponding coefficient ring must be stated. If q is even, inverting q also inverts two and removes the reflected two-torsion. For odd q , every p^{1-n} is one modulo two, and the ordinary maximal multiplicity persists over $\mathbb{Z}[1/q]$. These are statements about the chosen formal coefficient lattice, not about independence of the evaluated periods.

8 Polylogarithm identities and pole-cancelled order derivatives

Put $\zeta_q = e^{2\pi i/q}$ and initially define $\mathrm{Li}_s(z) = \sum_{n \geq 1} z^n / n^s$ for the roots considered here and $\Re s > 1$. The standard continuation conventions are those of [7, 6]. Absolute convergence gives

$$\sum_{py=x} \mathrm{Li}_s(e^{2\pi iy}) = p^{1-s} \mathrm{Li}_s(e^{2\pi ix}). \quad (8.1)$$

Indeed the sum of p -th roots of unity filters the summation index to multiples of p .

For a fixed rational angle, splitting the series into residue classes also gives

$$\mathrm{Li}_s(\zeta_q^a) = q^{-s} \sum_{r=1}^q \zeta_q^{ar} \zeta(s, r/q). \quad (8.2)$$

The Hurwitz-zeta terms have only a common simple pole at $s = 1$, of residue one. For $a \not\equiv 0 \pmod{q}$, their root-of-unity coefficients sum to zero. Thus the left side is entire as a function of the order s at a nontrivial root; for $a = 0$ it is $\zeta(s)$. This also provides a direct continuation argument for all identities below.

Theorem 8.1 (Fixed-basis identities for every complex order). *For every grid index a , the compiler polynomials satisfy the meromorphic identity*

$$\mathrm{Li}_s(\zeta_q^a) = \sum_{b \in q\mathcal{B}_q} P_{a,b}((p^{1-s})_{p|q}) \mathrm{Li}_s(\zeta_q^b). \quad (8.3)$$

Here $q\mathcal{B}_q$ denotes the numerator representatives of the basis points. The coefficient functions are finite exponential polynomials in s . There are no exceptional weight denominators in this basis.

Proof. Every polynomial certificate is a linear combination of raw rows. Evaluate it with $A_p = p^{1-s}$ and use (8.1) in the absolutely convergent half-plane. Both sides continue meromorphically by (8.2), so the identity theorem gives (8.3) for every complex s . $\square$

The theorem is an identity theorem for a fixed finite presentation, not an assertion that these $\varphi(q)$ evaluated functions or any of their special values are a minimal numerical basis.

8.1 Two explicit all-order identities with short raw certificates

At level twelve write $F_a(s) = \mathrm{Li}_s(\zeta_{12}^a)$. The normal form of F_1 yields

$$\boxed{\begin{aligned} &\mathrm{Li}_s(e^{i\pi/6}) + \mathrm{Li}_s(e^{5i\pi/6}) + (1 + 3^{1-s}) \mathrm{Li}_s(-i) \\ &= (12^{1-s} - 6^{1-s})\zeta(s). \end{aligned}} \quad (8.4)$$

In formal notation, with $\alpha = A_2$ and $\beta = A_3$, its certificate is the sum of just three raw rows:

$$r_{3,1/4} + \beta r_{2,1/2} + \alpha \beta r_{2,0}. \quad (8.5)$$

Expanding gives $e_{1/12} + e_{5/12} + (1 + \beta)e_{3/4} - (\alpha^2\beta - \alpha\beta)e_0$, as required. Equivalently, ordinary triplication and duplication prove the same formula. This familiar origin is not disguised as an independent new functional equation; its role here is an all-order, pole-aware integration test with an exact certificate.

At level thirty let $F_a(s) = \mathrm{Li}_s(\zeta_{30}^a)$, and put $\alpha = 2^{1-s}$, $\beta = 3^{1-s}$, $\gamma = 5^{1-s}$. A less compact but useful generated identity is

$$\begin{aligned} F_1 &= (\beta\gamma - 1)\zeta(s) + (\alpha - 1)F_2 - F_8 - F_{12} - F_{14} \\ &\quad - (\beta + 1)F_{18} - (\gamma + 1)F_{20} - F_{24}. \end{aligned} \quad (8.6)$$

It too has a short raw proof. Before specialization, the residual of (8.6) is exactly

$$r_{2,1/15} - r_{3,3/5} + r_{5,1/3} + A_5 r_{3,0} + r_{5,0}. \quad (8.7)$$

For example, the first two rows cancel $e_{16/30}$, the next two cancel $e_{26/30}$ and $e_{10/30}$, and the last cancels $e_{6/30}$. The resulting endpoint coefficient is $1 - A_3 A_5$. Expanding these five rows is a complete symbolic proof; no relation search is needed.

Both formulas specialize to identities for all positive integer weights, to meromorphic values of the order, and, after cancellation, to every order derivative at one. Appendix A gives the entire level-twelve polynomial normal-form table. The supplied JSON certificates contain all point identities at seven levels.

8.2 The endpoint coefficient vanishes at ordinary weights

For a nonzero index a ,

$$P_{a,0}(1, \dots, 1) = 0. \quad (8.8)$$

There is a purely algebraic reason. At $A_p = 1$, the zero-coordinate coefficient in every row vanishes: for parent zero the two copies of e_0 cancel, and for nonzero parent no zero preimage occurs. Therefore every certified relation has zero coefficient on e_0 . Applying this to $e_a - \text{NF}(a)$ proves (8.8). The removable pole in (8.3) is consequently certified before any numerical evaluation.

Theorem 8.2 (All-order finite-part rule). *Let $a \neq 0$, and define*

$$C_{a,b}(t) = P_{a,b}((p^{-t})_p), \quad L_b^{(j)} = \left. \frac{d^j}{ds^j} \text{Li}_s(\zeta_q^b) \right|_{s=1} \quad (b \neq 0).$$

Use $\zeta(1+t) = t^{-1} + \sum_{k \geq 0} (-1)^k \gamma_k t^k / k!$. For every $j \geq 0$,

$$\begin{aligned} L_a^{(j)} = & \sum_{\substack{b \in q\mathcal{B}_q \\ b \neq 0}} \sum_{h=0}^j \binom{j}{h} C_{a,b}^{(h)}(0) L_b^{(j-h)} + \frac{C_{a,0}^{(j+1)}(0)}{j+1} \\ & + \sum_{h=1}^j \binom{j}{h} C_{a,0}^{(h)}(0) (-1)^{j-h} \gamma_{j-h}. \end{aligned} \quad (8.9)$$

Coefficient derivatives are explicit. A monomial $c \prod_p A_p^{m_p}$ contributes $c(-\sum_p m_p \log p)^h$ to the h -th derivative, with the usual order-zero convention.

Proof. Substitute $s = 1 + t$ in (8.3). Equation (8.8) gives $C_{a,0}(0) = 0$. Multiply its Taylor series by the displayed Laurent series of zeta. The pole term contributes $C_{a,0}^{(j+1)}(0)/(j+1)$ after taking j derivatives; the regular zeta coefficients give the last sum. The nonendpoint terms follow by Leibniz's rule. All series converge in a sufficiently small punctured neighborhood, with a removable singularity in the product, so coefficient extraction is legitimate. $\square$

The derivative of order $j+1$ in the pole contribution is indispensable. Replacing a removable product by a naive value-times-value expression would lose this term.

For (8.4), put

$$D_h = (-\log 12)^h - (-\log 6)^h, \quad F_a^{[j]} = \left. \partial_s^j \text{Li}_s(\zeta_{12}^a) \right|_{s=1}.$$

The complete derivative family becomes

$$\begin{aligned} F_1^{[j]} + F_5^{[j]} + F_9^{[j]} + \sum_{h=0}^j \binom{j}{h} (-\log 3)^h F_9^{[j-h]} = & \frac{D_{j+1}}{j+1} \\ & + \sum_{h=1}^j \binom{j}{h} D_h (-1)^{j-h} \gamma_{j-h}. \end{aligned} \quad (8.10)$$

In particular,

$$F_1^{[0]} + F_5^{[0]} + 2F_9^{[0]} = -\log 2, \quad (8.11)$$

$$F_1^{[1]} + F_5^{[1]} + 2F_9^{[1]} - \log 3 F_9^{[0]} = \frac{\log^2 12 - \log^2 6}{2} - \gamma_0 \log 2. \quad (8.12)$$

For order two the right side is

$$\frac{-\log^3 12 + \log^3 6}{3} + \gamma_0(\log^2 12 - \log^2 6) + 2\gamma_1 \log 2,$$

and the left side is $F_1^{[2]} + F_5^{[2]} + 2F_9^{[2]} - 2\log 3 F_9^{[1]} + \log^2 3 F_9^{[0]}$.

8.3 Two exact cyclotomic product identities

Using $\text{Li}_1(z) = -\text{Log}(1-z)$ with the branch obtained from the defining disk and then continued to the indicated roots, (8.11) gives

$$(1 - \zeta_{12})(1 - \zeta_{12}^5)(1 - \zeta_{12}^9)^2 = 2. \quad (8.13)$$

The pole-cancelled value of (8.6) gives

$$\begin{aligned} & (1 - \zeta_{30})(1 - \zeta_{30}^8)(1 - \zeta_{30}^{12})(1 - \zeta_{30}^{14}) \\ & \cdot (1 - \zeta_{30}^{18})^2(1 - \zeta_{30}^{20})^2(1 - \zeta_{30}^{24}) = 15. \end{aligned} \quad (8.14)$$

Exponentiation of the exact logarithmic identities proves these products without a branch ambiguity. Conversely, the product alone would not fix the logarithmic identity's additive multiple of $2\pi i$. As an independent algebraic check, the package gives integer polynomial quotients verifying that each product polynomial minus its right side is divisible by the corresponding cyclotomic polynomial.

8.4 Stable numerical evaluation of the derivative identities

Write the generalized Stieltjes expansion

$$\zeta(1+t, x) = t^{-1} + \sum_{k \geq 0} \frac{(-1)^k \gamma_k(x)}{k!} t^k.$$

Cancel the common pole in (8.2) *symbolically*. For $a \neq 0$ one obtains the finite identity

$$\partial_s^j \text{Li}_s(\zeta_q^a) \Big|_{s=1} = \frac{(-1)^j}{q} \sum_{k=0}^j \binom{j}{k} (\log q)^{j-k} \sum_{r=1}^q \zeta_q^{ar} \gamma_k(r/q). \quad (8.15)$$

This follows by the same Laurent multiplication and is used in the numerical script. It avoids evaluating a pole and subtracting it from nearby values numerically. The numerical residuals remain diagnostics, not proofs of the identities.

9 Executable evidence and verification boundaries

The proofs above are uniform in the level and jet length. The following finite tests were actually run for the delivered package; the JSON records contain individual cases, not just these totals.

Exact check	Completed instances
Polynomial normal forms, levels $2 \leq q \leq 120$	119 levels
Raw polynomial rows annihilated on that range	3,317
Point normal forms on that range	7,259
Independent certificate files (levels 12, 15, 30, 60, 72, 105, 120)	7
Point identities independently replayed in those files	414
Raw rows independently replayed in those files	366
Scalar parity patterns ($q \leq 500$, plus 840, 1260, 2310)	2,348
Binary raw-row checks in the scalar sweep	488,954
Finite binary-jet cases	1,635
Direct raw reflected Smith computations	136
Direct raw unreflected Smith computations	68
Resolution cases over fields of sizes 2, 3, 5	498
Short displayed-identity raw-row certificates	2
Exact cyclotomic product polynomial divisions	2
Deliberately corrupted certificates rejected	3

The scalar sweep includes *every* binary prime-weight assignment at each stated level. The jet sweep includes every binary coefficient assignment of lengths one, two, and three at levels 3, 4, 6, 9, 12, 15, 20, 24, 30, 60, and five further specified examples. These finite ranges are not extrapolated as a substitute for the proofs.

The core generator and independent certificate verifier use only Python integer arithmetic and the standard library. The direct Smith script uses SymPy 1.14.0 and constructs the raw matrices without importing the normal-form generator. The resolution checker likewise constructs the complex directly from its rational grid labels. It checks $d^2 = 0$, exactness in negative degrees over the tested fields, and the nonfixed-support property of Lemma 5.2.

The numerical script uses mpmath and checks twenty cases at 70 and 110 decimal working digits: the level-12 and level-30 identities at orders 2, $5/2$, and $0.4 + 0.7i$, plus the level-12 derivative family through order three at each precision. The ordinary-order left sides use the finite Hurwitz Fourier sum, while the normal-form right sides use polylogarithms. Derivatives use (8.15). These computations use no interval arithmetic; the saved residuals are not certified error bounds.

9.1 Trust boundary of the artifacts

The small independent verifier establishes the delivered polynomial identities from reconstructed raw rows. Analytic validity of the raw rows is proved in (8.1); it is not an assumption hidden in the verifier. The general homological results have written proofs and finite implementation tests, not Lean or other proof-assistant certificates. A checksum protects file integrity, not mathematical validity. No repository files were changed during this work.

10 Audit corrections and proposed integration

The existing characteristic-zero distribution theorem is not contradicted by the results here. It is strengthened to an integral statement and supplemented with reflection data. The following targeted changes address wording still present in the pinned manuscript.

10.1 Clausen weight parity versus conjugation parity

In Chapter 2, the section on fundamental constants says that odd Clausen values use odd characters. Under the chapter's declared convention, $\text{Cl}_{2m+1}(\theta) = \Re \text{Li}_{2m+1}(e^{i\theta})$, an odd *weight* Clausen value

is a cosine component and uses even characters. The unambiguous replacement is:

Sine-series, or conjugation-odd, components use odd characters. With $\text{Cl}_{2m} = \Im \text{Li}_{2m}$ and $\text{Cl}_{2m+1} = \Re \text{Li}_{2m+1}$, the odd-weight Clausen components instead use even characters.

This corrects a parity ambiguity without changing the subsequent weight-two Clausen formulas.

10.2 Failure of a reduction search is not nonreduction

The Family 2 discussion in Chapter 4 states that the higher-weight antisymmetric parts do not reduce against the $(i, 1)$ basis alone. The surrounding account concerns bounded integer-relation searches and formal relation matrices. The numerical statement should instead say that the reported search did not find those reductions. A formal nonmembership statement must name the particular row presentation; it must not be promoted to nonmembership of the evaluated periods in a numerical algebra. This is consistent with the careful qualifications already present elsewhere in the manuscript.

10.3 Fourier transport is not unimodular lattice transport

The manuscript's Fourier transform is exact over a cyclotomic field. Its inverse contains group-order denominators, so equal dimensions and transported vector-space relations do not by themselves identify Smith forms of integer lattices. An integral use of that discussion should specify the coefficient ring and any localization. Theorem 2.1 provides a separate row-row argument that avoids this obstruction for the complete distribution presentation. This is a clarification of the scope of Fourier transport, not a claimed error in the Fourier identity.

10.4 Updating the research-status paragraph

The scalar integral and modular questions, the reflected scalar Smith forms, and the one-variable integral-jet Smith forms can now be marked proved *for this complete presentation*. Multivariate reflection is reduced to an explicit Koszul homology problem, not to the one-variable minimum-valuation formula. The full equivariant module classification and numerical period-independence questions remain separate. The integration fragment uses the label prefix `intdist:` and is intended to accompany this report rather than silently replace the earlier characteristic-zero proof.

11 Further research questions

Equivariant integral module structure. Determine the full $R[J]$ -module structure of the universal quotient and of its reflected quotients, beyond Tate cohomology and abelian Smith forms. The explicit jet example (7.3) shows that even the torsion-free part has integral extension data that the Smith factors miss. A useful first target is a classification of these extensions at prime-power levels.

Multivariate jet torsion. For a specified Artinian algebra $\mathbb{F}_2[u_1, \dots, u_d]/\mathfrak{a}$, compute the Koszul homology of the binary parameters in closed form or from a minimal resolution of their ideal. Theorem 6.2 gives the exact target. Any proposed replacement for the minimum valuation must account for nonprincipal ideals and their syzygies.

Optimizing the original-symbol basis. The forbidden lift in each local fiber, and the forbidden prime chosen at a point, can be varied. Which choices minimize normal-form coefficient height, support, or certificate length while retaining unit triangularity? The present height bound is uniform but intentionally coarse.

Integral Fourier and character interfaces. Construct explicit comparison matrices between the original-symbol basis, conductor-character coordinates, and primitive grids over the smallest useful localized cyclotomic integer rings. The resulting indices should separate coordinate singularities from actual torsion and make integral transport between polylogarithm and Hurwitz-zeta certificates auditable.

Higher-depth distributions. Extend the division-free certificate interface to depth-two distribution relations together with specified shuffle, stuffle, and regularization rows. Such a module is not the depth-one module treated here. The surviving S_6 question requires additional analysis in that setting; it is not resolved by reinterpreting a depth-one Smith computation.

Analytic jets with certified errors. Combine the exact finite-part formulas with interval-certified Hurwitz or generalized Stieltjes evaluation, including explicit cancellation budgets. This would turn the current diagnostics into certified numerical enclosures while leaving equality proofs in the exact polynomial layer.

Proof-assistant formalization. The shortest initial targets are the selector lemma, termination of (4.1), exact back-substitution, and the matrix base-change theorem. A second stage can formalize the filtered resolution and the acyclic-subcomplex proof of Theorem 6.2. The analytic continuation layer can then be attached independently.

Arithmetic evaluation kernels. Determine which relations among evaluated cyclotomic periods are not visible in the declared distribution presentation. A formal basis or nonzero formal class is not an independence theorem for numbers. Any claim about the evaluation kernel should state its weight, coefficient field, and available functional equations explicitly.

A The complete level-twelve normal-form table

Let $E_a = [e_{a/12}]$, $\alpha = A_2$, $\beta = A_3$. The retained basis is E_0, E_5, E_8, E_9 . Each row gives the four coefficients of E_a in that basis.

a	E_0	E_5	E_8	E_9
0	1	0	0	0
1	$\alpha\beta(\alpha - 1)$	-1	0	$-\beta - 1$
2	$\alpha(\beta - 1)$	0	$-\alpha - 1$	0
3	$\alpha(\alpha - 1)$	0	0	-1
4	$\beta - 1$	0	-1	0
5	0	1	0	0
6	$\alpha - 1$	0	0	0
7	$\alpha(\beta - \alpha)$	1	$-\alpha(\alpha + 1)$	$\beta + 1$
8	0	0	1	0
9	0	0	0	1
10	$1 - \beta$	0	$\alpha + 1$	0
11	$\alpha(1 - \beta)$	-1	$\alpha(\alpha + 1)$	0

The companion file `data/normal_forms_q12.json` contains the unfactored integer monomial data. At arbitrary complex order, substitute $\alpha = 2^{1-s}$, $\beta = 3^{1-s}$ and $E_a = \text{Li}_s(e^{2\pi ia/12})$. At formal integral jet weights, substitute polynomials modulo u^L instead. The same table supports both uses.

B Reproduction and artifact map

From the package root, the standard-library checks are

```
python code/run_checks.py
python code/check_resolution.py
python code/check_short_identities.py
python code/verify_certificates.py data/normal_forms_q12.json
```

The optional independent matrix and numerical checks are

```
python -m pip install -r requirements-optional.txt
python code/check_smith.py
python code/check_products_and_controls.py
python code/numerical_checks.py
```

The optional dependencies are SymPy and mpmath; the core verifier needs neither. Build the article with `python code/build_article.py` or run pdfLaTeX twice in the `article` directory. Every data file names its scope. In particular, `numerical_checks.json` explicitly states that it does not contain interval certificates.

The repository integration notes and focused corrections are in `integration/`. The source snapshot and read file-blob hashes are recorded in `PROVENANCE.json`. The checksum manifest is generated only after compilation and final verification. No external font files or third-party source PDFs are included.

References

- [1] V. Reshetnikov, ProveIt project, *Polylogarithms and their Arithmetic Bridges*, manuscript and research-status chapters, repository snapshot 28357e8ca63dd78327db91d9be239d75e4462879. <https://github.com/VladimirReshetnikov/ProveIt/tree/28357e8ca63dd78327db91d9be239d75e4462879/Analysis/Polylogarithms/docs/manuscript>.
- [2] ProveIt project, *Distribution jets* research report, `Analysis/Polylogarithms/docs/reports/distribution-jets/article/distribution_jets.tex`, repository snapshot as above; in particular the universal conductor normal form and its characteristic-zero hypothesis.
- [3] D. S. Kubert, The universal ordinary distribution, *Bulletin de la Société Mathématique de France* **107** (1979), 179–202. doi:10.24033/bsmf.1891.
- [4] Y. Ouyang, On the universal norm distribution, *Journal of the Ramanujan Mathematical Society* **17**, no. 4 (2002), 287–311. arXiv:math/0210297v2. The cited statements were checked in the arXiv version.
- [5] Y. Ouyang, Spectral sequences of universal distribution and Sinnott’s index formula, preprint (1999), arXiv:math/9911031v1. See the fixed-point double-complex argument and Theorem 7.1.
- [6] NIST Digital Library of Mathematical Functions, Section 25.11, Hurwitz Zeta Function. <https://dlmf.nist.gov/25.11>.
- [7] NIST Digital Library of Mathematical Functions, Section 25.12, Polylogarithms. <https://dlmf.nist.gov/25.12>.